Supporting Information

Mg^{2+}/Al^{3+} Co-doped Li-Rich Manganese-based Oxides for Boosting Rate Performance and Stability of Lithium-Ion Batteries

*Junxia Meng[a], Wenzhuan Hu[a], Quanxin Ma[b,c*1], Zhenzhen Wu[d], Lishuang Xu[a], Jie Huang[a], Chen Xiao [a], Junru Wang[a], Lina Zhang[a], Feng Liu[a], Xing Zhi[e,f*2], Shanqing Zhang[d,e,f*3]*

[a]School of Physics and Electronics, Gannan Normal University, Ganzhou, 341000, China

[b]Jiangxi Provincial Key Laboratory of Power Batteries & Energy Storage Materials, School of Materials Science and Engineering, Jiangxi University of Science and Technology, Ganzhou, 341000, China

[c]Yichun Lithium New Energy Industry Research Institute, Jiangxi University of Science and Technology, Yichun, 336000, China

[d]School of Environment and Science, Gold Coast Campus, Griffith University, QLD 4222, Australia

[e]Institute for Sustainable Transformation, School of Chemical Engineering and Light Industry, Guangdong University of Technology, Guangzhou, 510006 China

[f]Guangdong Provincial Laboratory of Chemistry and Fine Chemical Engineering Jieyang Center, Jieyang, 515200 China

1.Experimental Sections

1.1 Materials synthesis

***Synthesis of cathode materials**:* In this experiment, $Li_{1.2}Mn_{0.56}Ni_{0.16}Co_{0.08}O_2$ was prepared by coprecipitation method and high solid-phase sintering process as follow: Stoichiometric amount of $MnSO_4 \cdot H_2O$, $NiSO_4 \cdot 6H_2O$ and $CoSO_4 \cdot 7H_2O$ with corresponding molar ratio were dissolved in deionized water to achieve a 2 mol L^{-1} mixed solution, and 2 mol L^{-1} NaOH with the desired amount of $NH_3 \cdot H_2O$. Both solutions were slowly add into the reaction kettle in a nitrogen atmosphere. After the reaction is complete. The precursor powder was filtered, washed, and then dried at 100 °C for 12 h, finally obtained the hydroxide precursor. The stoichiometric amounts of Li_2CO_3 were mixed with the as prepared precursor, and ball milling for 10 h. The mixtures powder after ball milling was heat-treated at 450 °C for 6 h. After that, the temperature was raised to 950°C for 16 h to obtained the $Li_{1.2}Mn_{0.56}Ni_{0.16}Co_{0.08}O_2$, which was labeled as LRMO. Similarly, the Mg-doped, Al-doped and Mg/Al co-doping $Li_{1.2}Mn_{0.56}Ni_{0.16}Co_{0.08}O_2$ samples are obtained by the similar process by adding moderate content $MgSO_4$, $Al(NO_3)_3 \cdot 6H_2O$ and both $MgSO_4$ and $Al(NO_3)_3 \cdot 6H_2O$ together, here denoted as Mg-LRMO, Al-LRMO and Mg/Al-LRMO, respectively.

***Pouch Full Cell Preparation**:* A high-quality cathode paste was prepared by mixing the active material, conductive agent (acetylene black), and binder (PVDF) in a ball mill tank at a mass ratio of 90:5:5. The paste was uniformly coated onto an aluminum foil collector with a surface density of approximately 24 mg cm^{-2}. The coated cathode sheet was vacuum-dried at 120 °C for 12 hours, then cut and rolled. Each cathode measured 3.6×50 cm. Similarly, the anode was prepared by mixing graphite, conductive agent (acetylene black), CMC (carboxymethyl cellulose), and styrene-butadiene rubber at a mass ratio of 94:2:2:2 to form a slurry. This slurry was uniformly coated onto a copper foil with a surface density of 18 mg cm^{-2}. The N/P ratio of the full battery was 1.05. Each anode was cut to a size of 4.0×55 cm. The cathode, separator, and anode were stacked in sequence and wound into a 4×3×0.48 cm cell. After drying at 100 °C overnight, the cell was assembled and filled with a commercial electrolyte in an argon-filled glovebox, resulting in the LRMO and Mg/Al-LRMO pouch cells.

1.2 Material Characterization

An inductively coupled plasma optical emission spectrometer (ICP-OES, Perkin Elmer Optima 5300 DV) is used to explore the metal content. X-ray diffractometer (XRD) was conducted to characterize the crystal structure of the synthesized materials with CuKα radiation. The diffraction data were collected in 2θ ranges of from 10° to 80°. Crystal structure analysis software (GSAS) is used to refine the structure and obtained the lattice parameters. Scanning electron microscopy (HITACHI, S-4700) were used to examined the morphology of materials with a working voltage of 3 KV, the energy dispersive X-ray spectrometer (EDS) were used to analyze the elemental distribution of Ni, Co, Mn, Mg and Al on the surface of powder. Transmission electron microscope (TEM) and a selective area electron diffraction (SAED) were used to observe the micromorphology and microstructure of materials. Modified fast Fourier transform (FFT) or inverse FFT (IFFT) was done using DigitalMicrograph software. X-ray photoelectron spectroscopy (XPS, PHI 5700 ESCA, USA) was utilized to determine the oxidation states of surface element.

1.3 Electrochemical Measurements

The preparation process of the half-cell is as follows: Coin-type cells were constructed in an argon-filled glovebox (<0.1 ppm O_2 and H_2O). The working electrodes were perpared by mixing the active materials, acetylene black, and polyvinylidene difluoride binder with a weight ratio of 8:1:1.

The half-cells were assembled with a positive electrode, a separator, a lithium plate, and an electrolyte. The electrolyte was 1.0 M $LiPF_6$ dissolved in ethyl carbonate/dimethyl carbonate/diethyl carbonate by volume 1:1:1. To obtain reliable measurement data, the loading mass of active material was *ca*. 2.28 mg cm^{-2}. The full cells were prepared by assembling the as-prepared electrodes with commercial graphite. The cycling and rate performances were measured on by a galvanostatical charge-discharge test (Neware, China). The test voltage range was 2.5-4.6 V vs Li/Li^+. The rate performance was evaluated at 0.1-5.0 C (1 C=250 mA g^{-1}). Electrochemical impedance spectroscopy (EIS) and Cyclic voltammetry (CV) measurement were tested on an electrochemical workstation (VMP3) in the frequency range of 0.01 Hz to 100 kHz `and at a scan rate at a scan rate from 0.2 to 1.0 mV s^{-1} between 2.5 and 4.6 V. All electrochemical tests are carried out at room temperature.

2. Supplementary methods

2.1 First-principles calculations.

All calculations were performed using spin-polarised density functional theory (DFT) implemented in the Vienna ab initio simulation package (VASP) code. [1] The generalized gradient approximation of Perdew, a spin-polarized version of generalized gradient approximation (GGA) with Perdew-Burke-Ernzerhof (PBE) functional was adopted to treat the exchange and correlation functional. [2] The valence electrons and charge density are described as plane waves with a cut-off energy of 500 eV. To describe correctly the energetics of the transition metal 3d states, Hubbard correction was introduced in our calculations. The *U* values of 3.9, 6.0, 3.5, 4.1 and 4.0 eV were used for Mn, Ni, Co, Mg and Al, respectively, according to previous literature. In our calculations, we used a 2×2×2 supercell of Li_2MnO_3 using a spacing smaller than 0.05˚ A^{-1}. The Monkhorst-Pack scheme 3×2×3 k-points mesh was used for the integration in the irreducible Brillouin zone. The atomic coordinates and lattice parameters were fully relaxed during the geometry optimization process. Convergence to self-consistent iterations was assumed when the total energy difference between cycles was less than 10^{-6} eV. The lattice parameters and the ionic position of the structure were fully relaxed, and the final forces on every relaxed atoms were less than 0.01 eV/Å. The total and projected density of states (TDOS/PDOS) and charge density distribution were further calculated for the optimized structures. The covalent and ionic bonding properties between metal and O ions were analyzed by examining the crystal orbital Hamilton population (COHP) using a VASP-internal subroutine and taken from the resulting output file. Local functions were constructed within VASP using Bessel functions, COHP analyses have been performed using LOBSTER code. [3]

2.2 Calculation of Li^+ diffusion coefficient.

2.2.1. CV test.

The D_{Li^+} measured by CV can be according to the Randles-Sevcik equation by fitting the relationship between the scan rate and the peak current (I_p) as follows (T=298 K). [4]

$$I_p^2=(2.686\times10^5)^2 n^3 A^2 C \nu D_{Li^+} \qquad \text{Equation S1}$$

where I_p is the peak current under different scan rates (A), n is the number of electrons transferred, A is the surface area of electrode material (cm^2), C is the molar concentration of Li^+ in the bulk electrode ($cm^2\ s^{-1}$), ν is the scan rate ($mV\ s^{-1}$) and D_{Li^+} is the diffusion coefficient of Li^+, respectively.

2. EIS test.

The diffusion coefficient of Li^+ (D_{Li^+}) can be calculated by the following formula:[5]

$$D_{Li^+} = 0{\cdot}5R^2T^2/n^4A^2F^4C^2\sigma^2 \qquad \text{Equation S2}$$

where R, T, A, F, n, C, and σ is the ideal gas constant, absolute temperature, surface area of the electrode, Faraday constant, number of electrons per molecule during oxidation, bulk concentration of Li^+ in the inactive material and Warburg impedance coefficient, respectively.

3. GITT test.

Galvanostatic intermittent titration technique (GITT), an effective electrochemical method to calculate the D_{Li^+} is employed in this work. By measuring various voltage increases or drops during the titration process and using the following equation, the D_{Li^+} can be obtained by Equation S3. [6]

$$D_{Li^+} = \frac{4}{\pi}\left(\frac{mV_M}{MS}\right)^2\left(\frac{\Delta Es}{\tau(\Delta E_\tau/d\sqrt{\tau})}\right)^2; (\tau \ll L^2/D_{Li^+}) \qquad \text{Equation S3}$$

where m refers to the mass of active material in the electrode, M represents the molecular weight of LMNCO material, V_M is the molar volume, L is the thickness of the electrode and S is the area of the electrode, τ is the time duration in which the current is applied, $E\tau$ and Es are the transient voltage change during the single titration current flux and the steady voltage change after the relaxation period, respectively.

If $E\tau$ and $\tau^{1/2}$ exhibits a linear relationship, Equation S3 can be further simplified as equation S4. [7]

$$D_{Li^+} = \frac{4}{\pi\tau}\left(\frac{\mathrm{mV_M}}{\mathrm{MS}}\right)^2\left(\frac{\Delta\mathrm{Es}}{\Delta\mathrm{E}_\tau}\right)^2; (\tau \ll \mathrm{L}^2/D_{Li^+}) \qquad \text{Equation S4}$$

Based on equation S4 and the GITT test result, the calculated $D_{Li}{}^{+}$ values at varied charge/discharge potentials were obtained.

4. Layer spacing

The refined lattice parameters were further employed to calculate the thickness of the Li slab. The slab thickness (S_{MO2}) and interslab thickness (I_{LiO2}) can be calculated based on the following formula:[8]

$$S_{(MO2)} = 2\ (1/3\text{-}Z_{ox})C \qquad \text{Equation S5}$$

$$I_{(LiO2)} = C/3\text{- } S_{(MO2)} \qquad \text{Equation S6}$$

In Equation S5 and S6, Z_{ox} is the coordinated 6c lattice position occupied by O^{2-} in the c-direction and C is the lattice constant (c) in *R-3m* structure.

5. Energy Density

The energy density (W) can be calculated by the following formula:

$$\mathrm{W} = \frac{\mathrm{CU}}{\mathrm{G}} \qquad \text{Equation S7}$$

where W refers to practical energy density, C represents the battery discharge capacity, U refers the average discharge voltage, G is the total battery mass.

Supplementary Figures

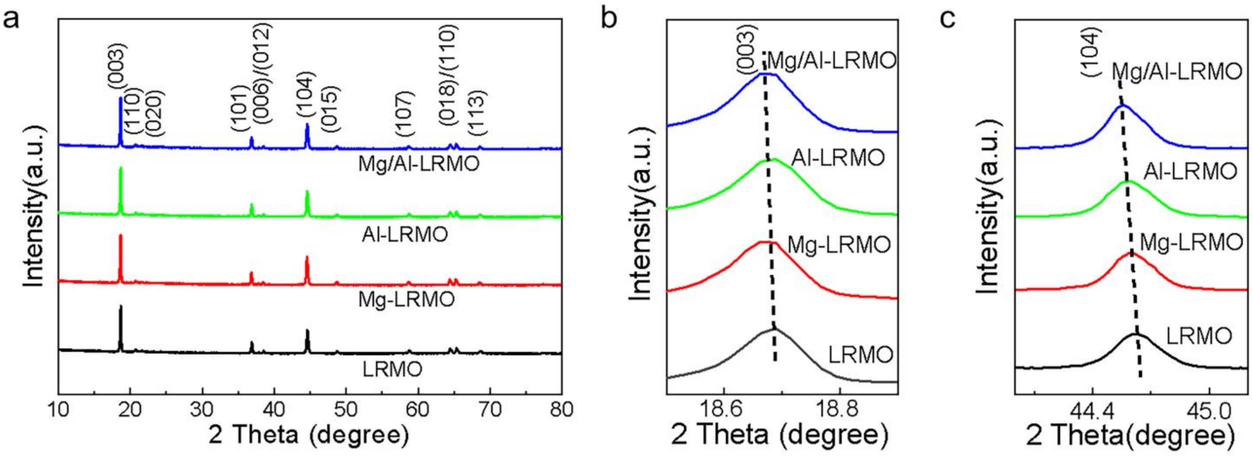

Figure S1. (a) XRD patterns (b) magnified (003) and (c) magnified (104) peaks of LRMO, Mg-LRMO, Al-LRMO and Mg/Al-LRMO.

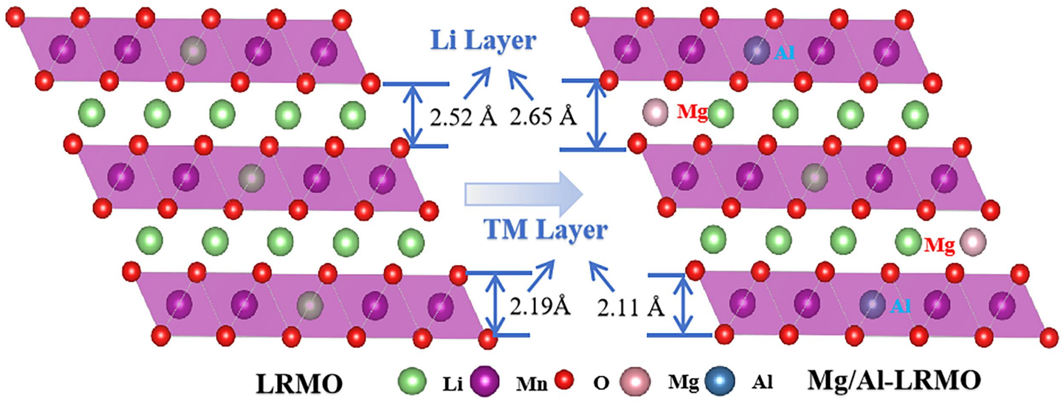

Figure S2. Structure diagram of LRMO and Mg/Al-LRMO.

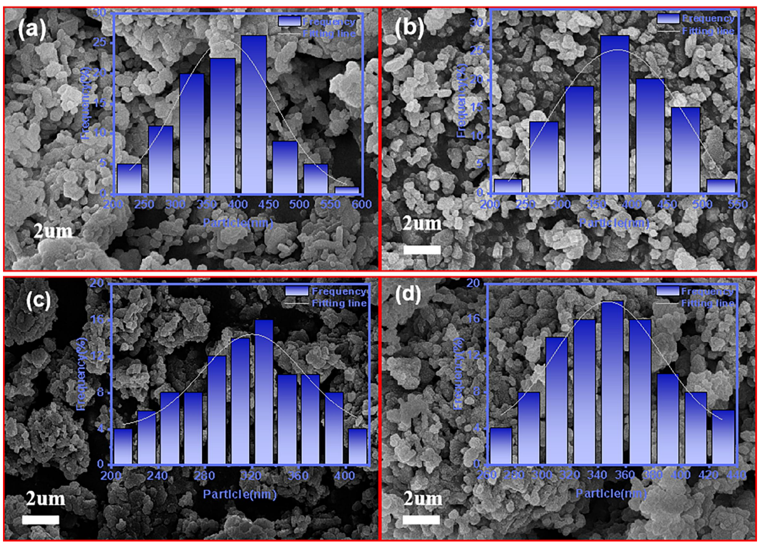

Figure S3. SEM images of (a) LRMO, (b) Mg-LRMO, (c) Al-LRMO and (d) Mg/Al-LRMO.

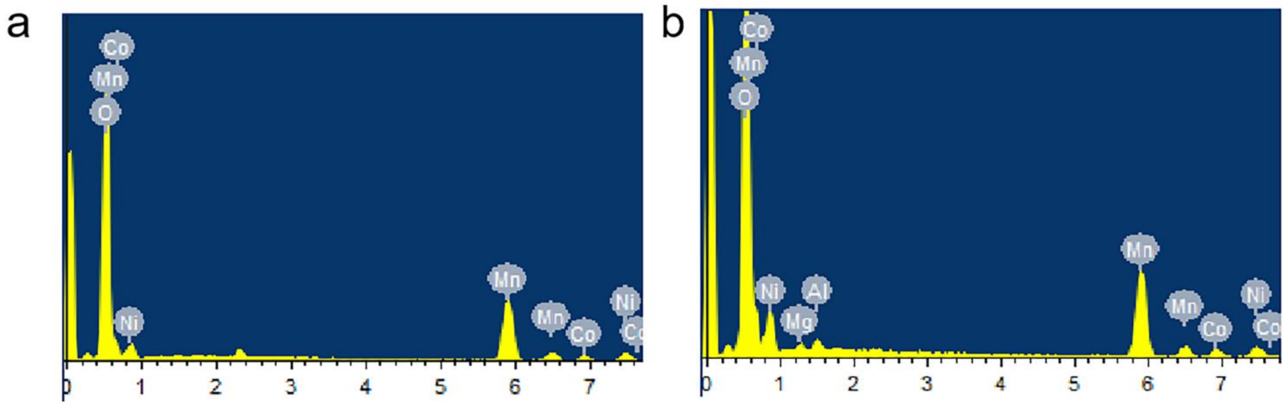

Figure S4. EDS of (a) LRMO and (b) Mg/Al-LRMO from SEM.

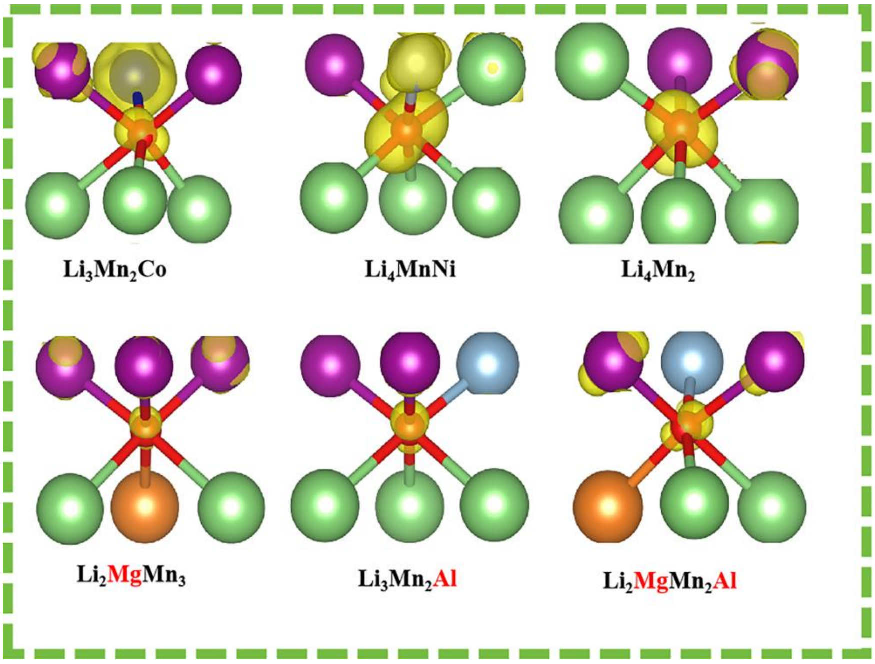

Figure S5. The highest occupied states (energy level of HOS is chosen as -1.5 to 0 eV) of O anions with different cation coordination.

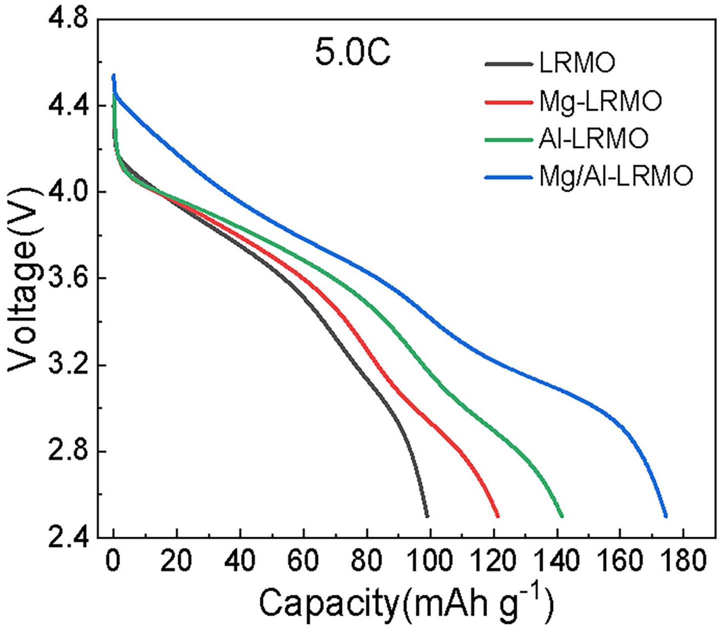

Figure S6. Discharge curves of LRMO, Mg-LRMO, Al-LRMO and Mg/Al-LRMO at 5.0 C, respectively.

Figure S7. The quality of LRMO and Mg/Al-LRMO pouch cells.

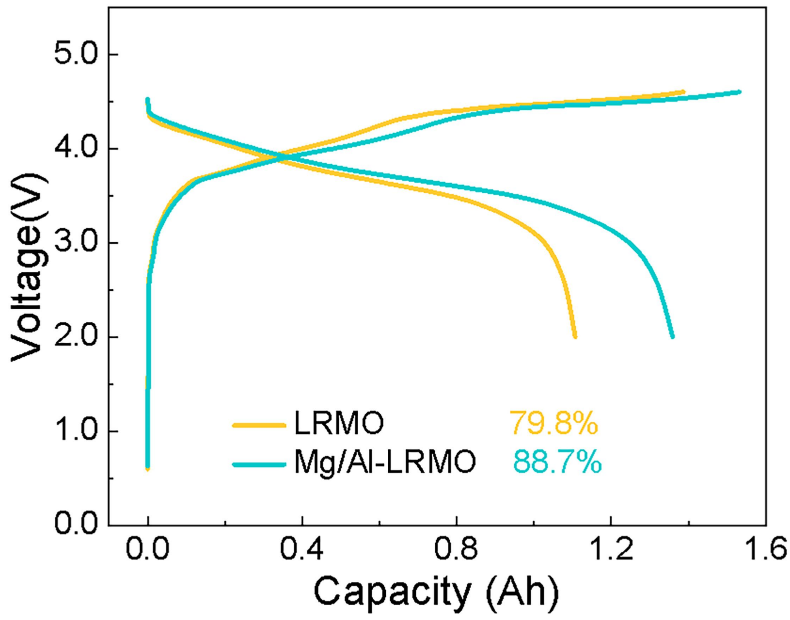

Figure S8. The initial charge/discharge profiles of LRMO and Mg/Al-LRMO pouch cells at 0.1 C.

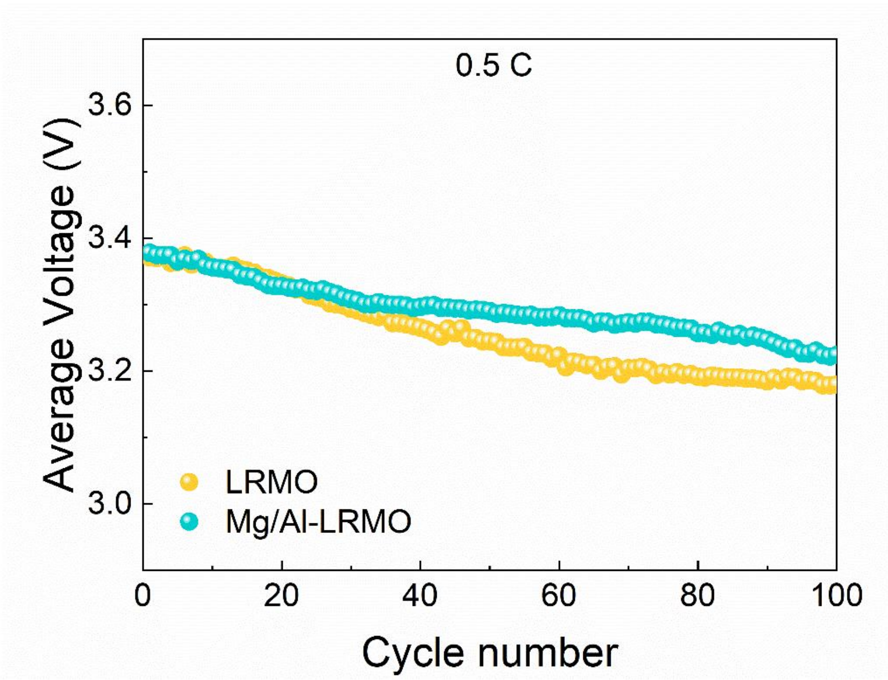

Figure S9. Average voltages of corresponding cycle of LRMO and Mg/Al-LRMO pouch cells at 0.5 C.

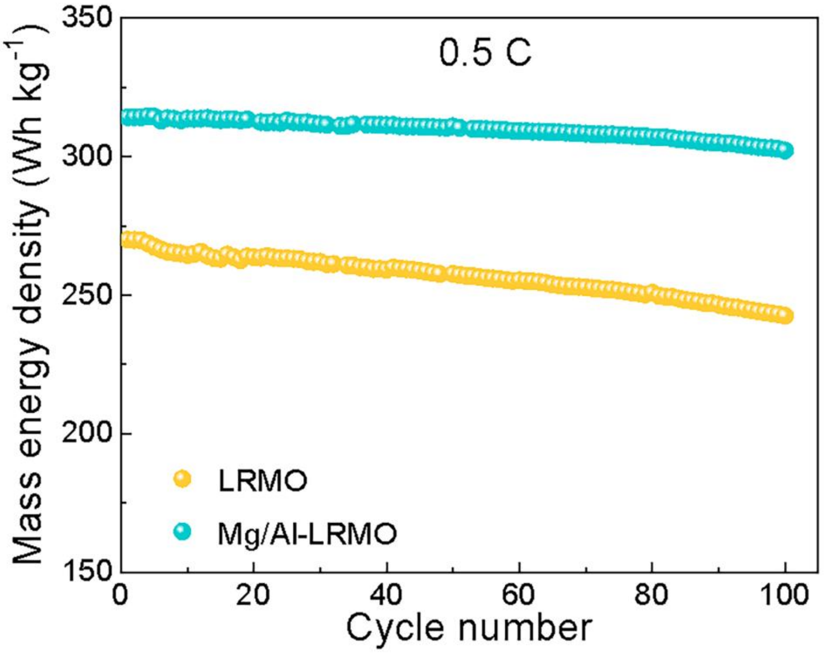

Figure S10. Mass energy density of LRMO and Mg/Al-LRMO pouch cells at 0.5 C.

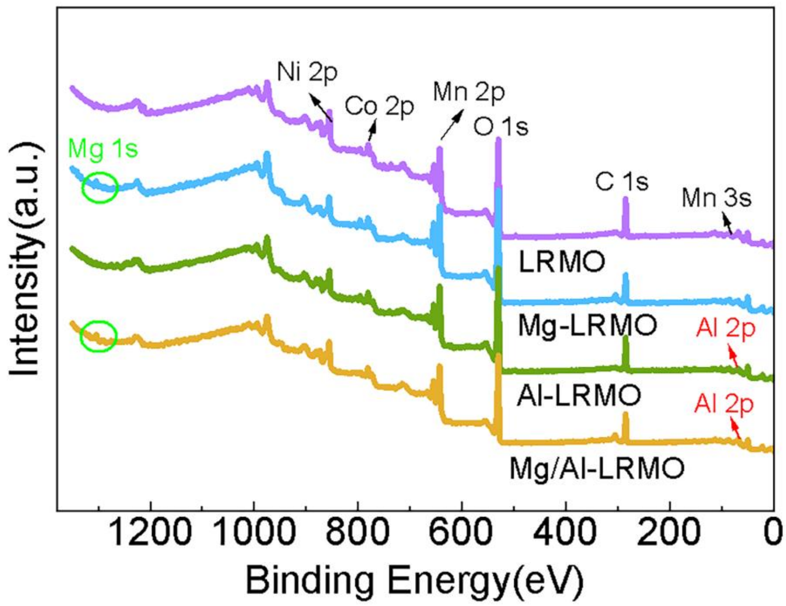

Figure S11. XPS survey spectrum for LRMO, Mg-LRMO, Al-LRMO and Mg/Al-LRMO.

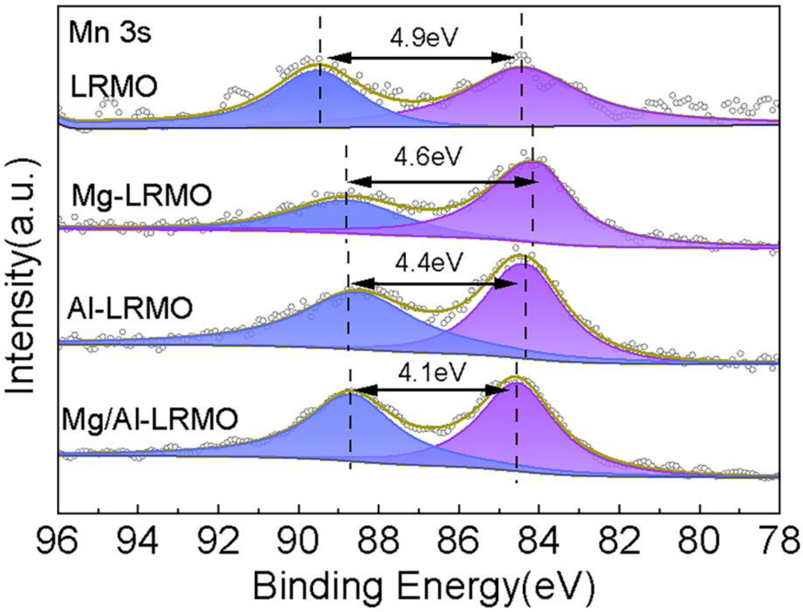

Figure S12. XPS spectra of Mn 3s for LRMO, Mg-LRMO, Al-LRMO and Mg/Al-LRMO.

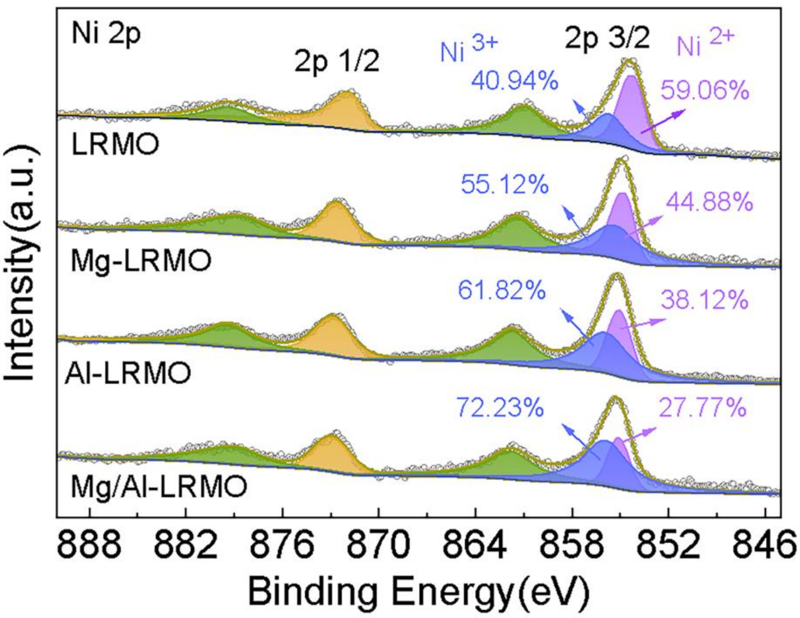

Figure S13. XPS spectra of Ni 2p for LRMO, Mg-LRMO, Al-LRMO and Mg/Al-LRMO.

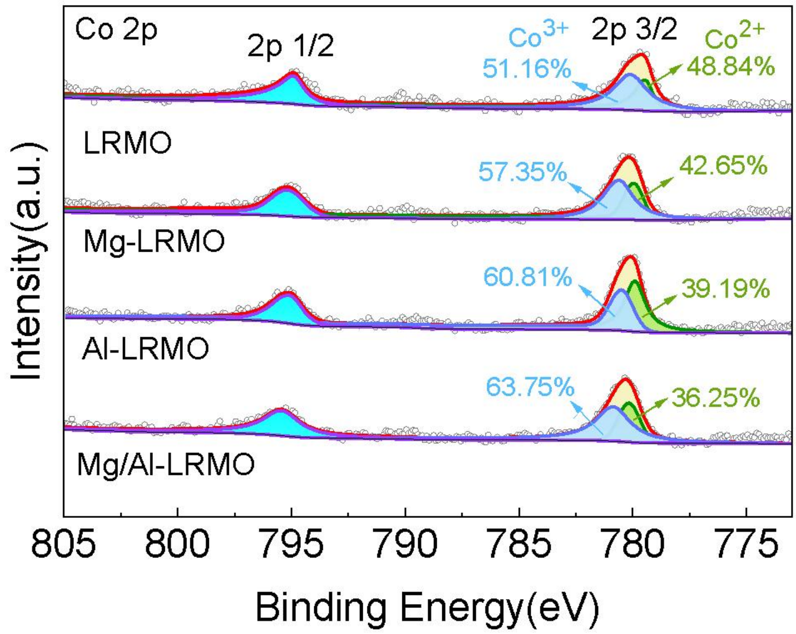

Figure S14. XPS spectra of Co 2p for LRMO, Mg-LRMO, Al-LRMO and Mg/Al-LRMO.

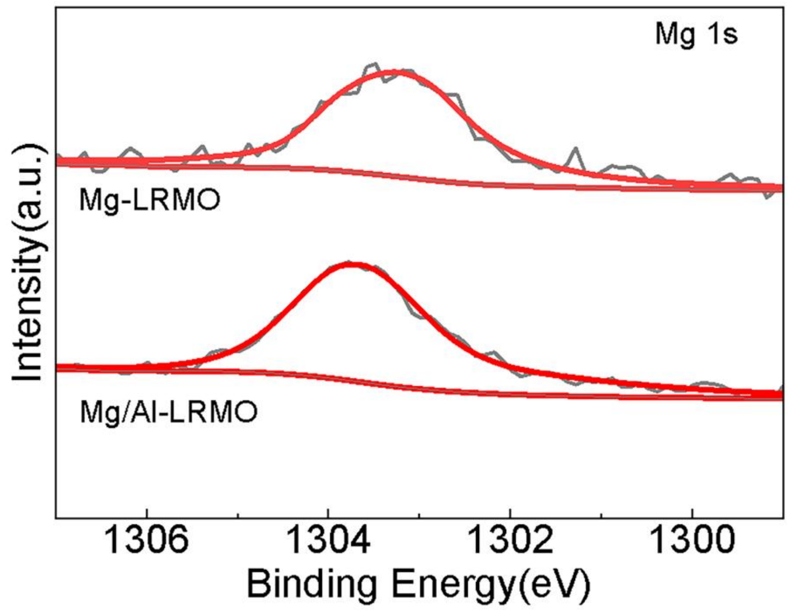

Figure S15. XPS spectra of Mg1s for Mg-LRMO and Mg/Al-LRMO.

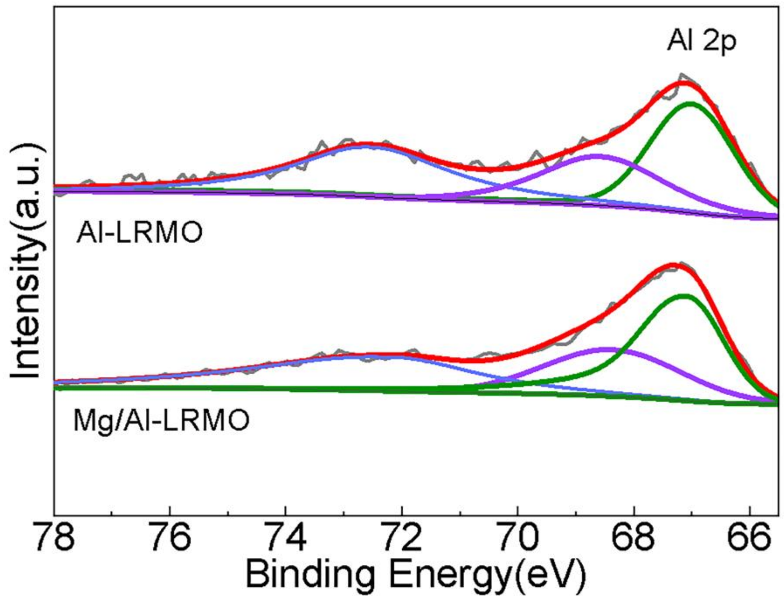

Figure S16. XPS spectra of Al 2p for Al-LRMO and Mg/Al-LRMO.

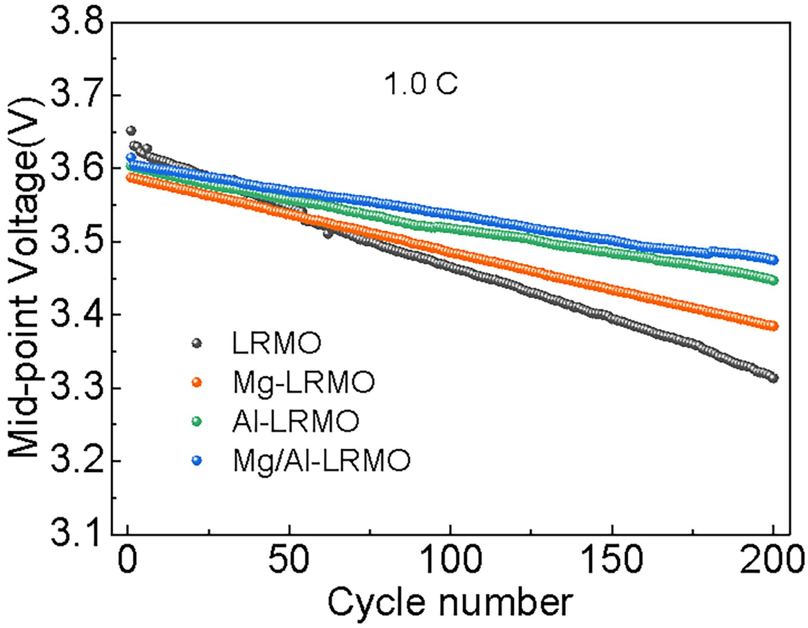

Figure S17. Mid-point voltage of LRMO, Mg-LRMO, Al-LRMO and Mg/Al-LRMO at 1.0 C.

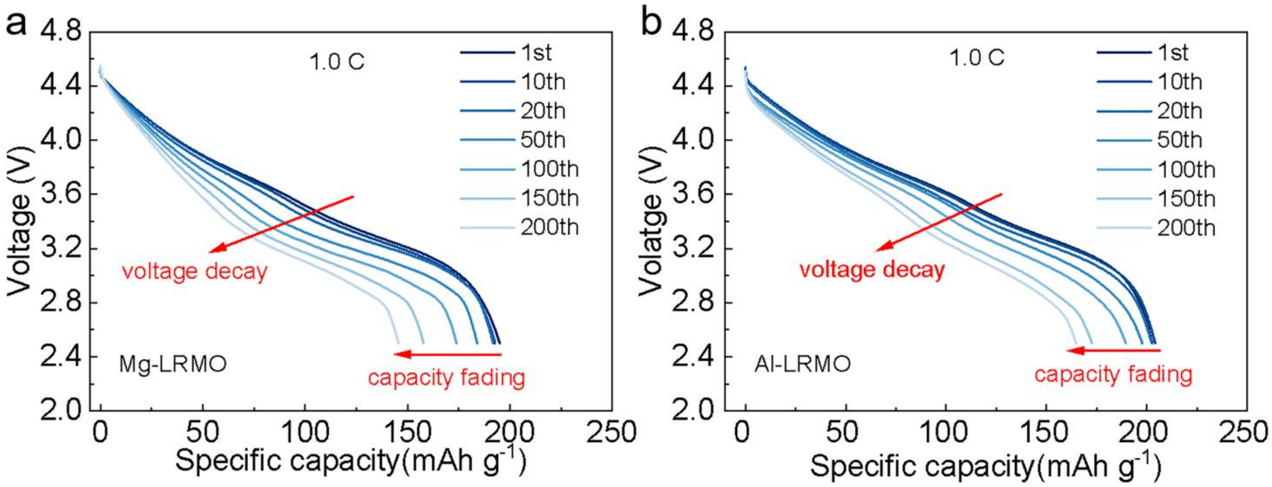

Figure S18. Discharge curves of (a) Mg-LRMO and (b) Al-LRMO at different cycles at 1.0 C.

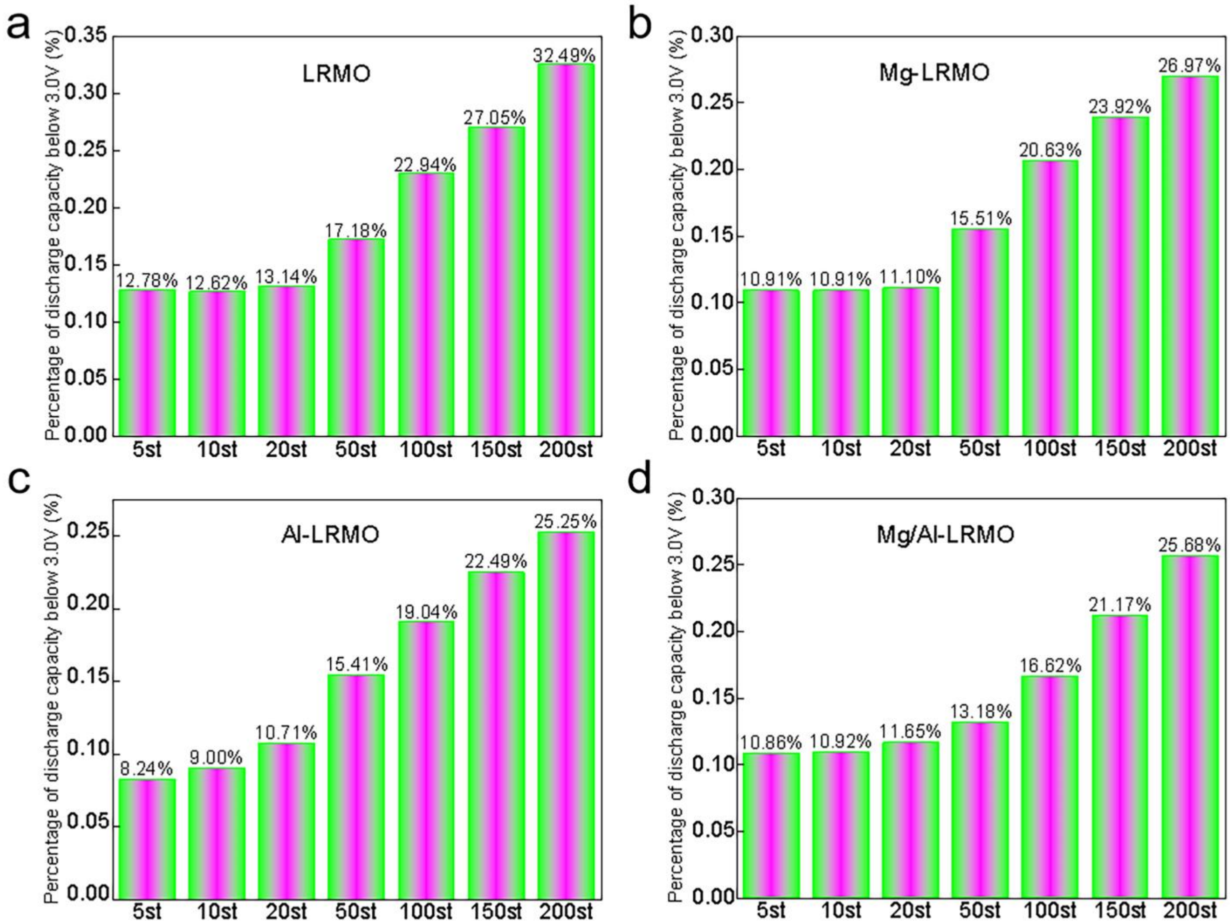

Figure S19. Percentage of discharge capacity below 3.0 V of (a) LRMO, (b) Mg-LRMO, (c) Al-LRMO and (d) Mg/Al-LRMO.

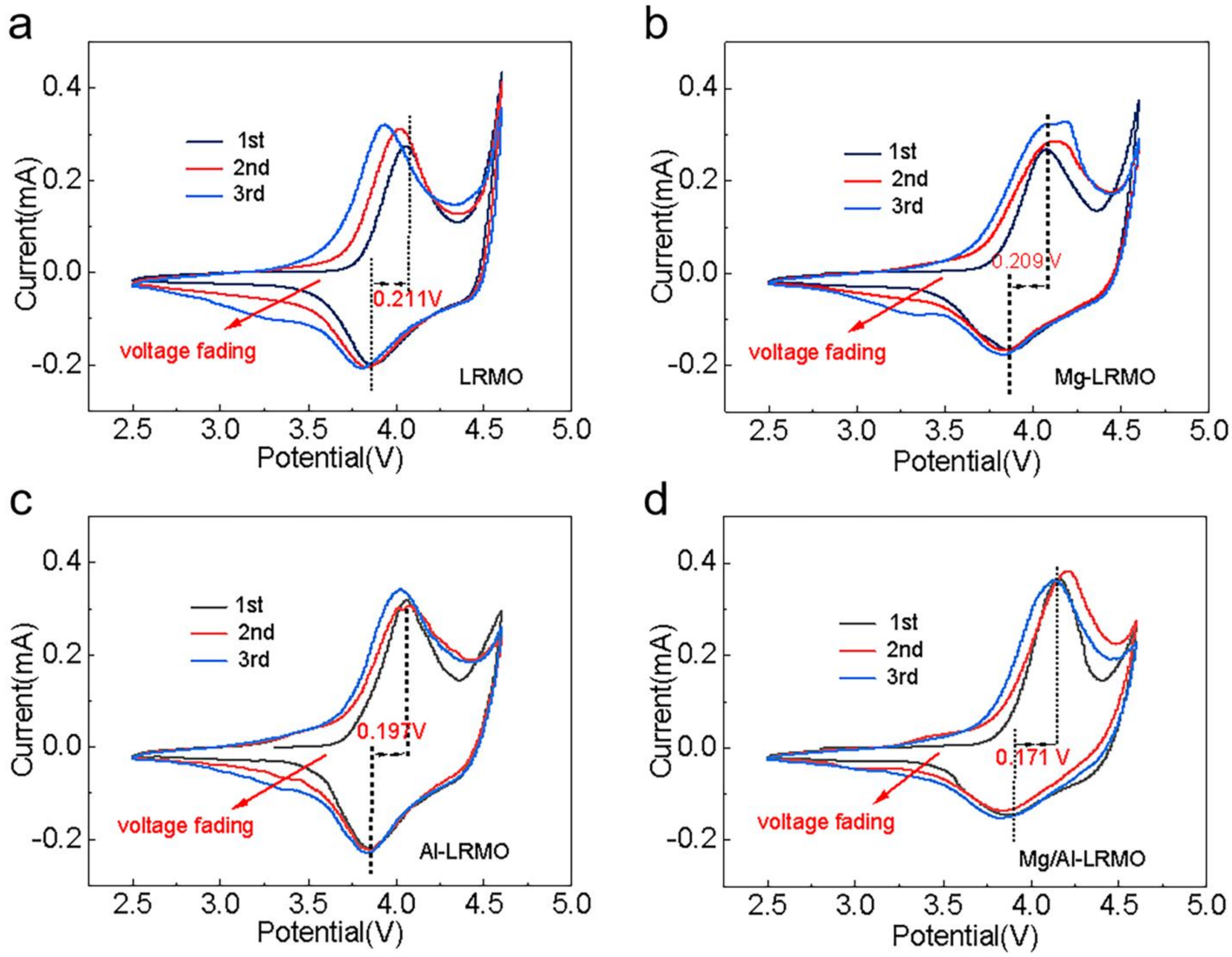

Figure S20. CV curves of (a) LRMO, (b) Mg-LRMO, (c) Al-LRMO, and (d) Mg/Al-LRMO at initial three cycles between 2.5 and 4.6 V at a scanning rate of $0.1 mVs^{-1}$.

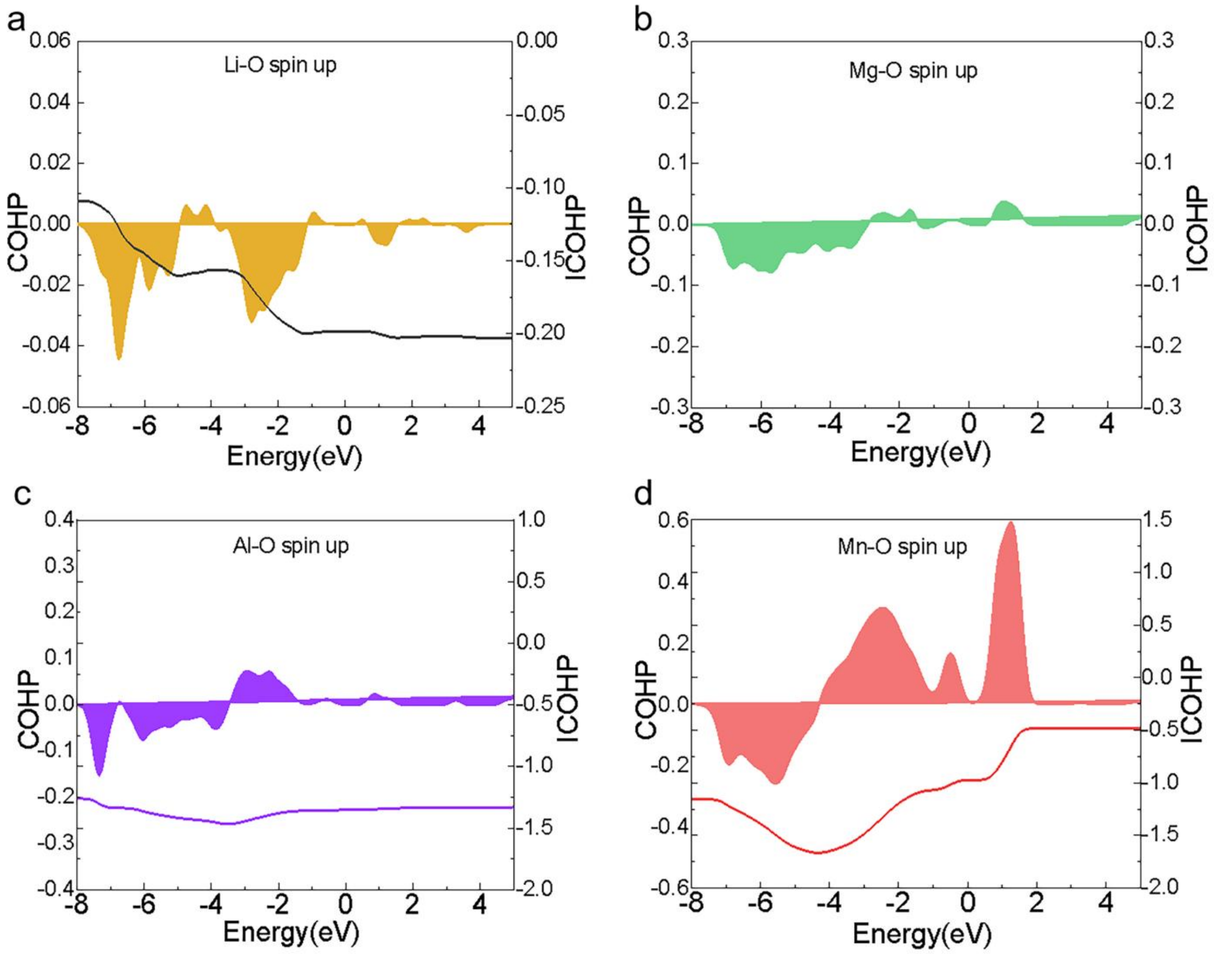

Figure S21. (a) The COHP and ICOHP analysis of Li-O (b) Mg-O, (c) Al-O, and (d) Mn-O bonds spin-up for Mg/Al-LRMO.

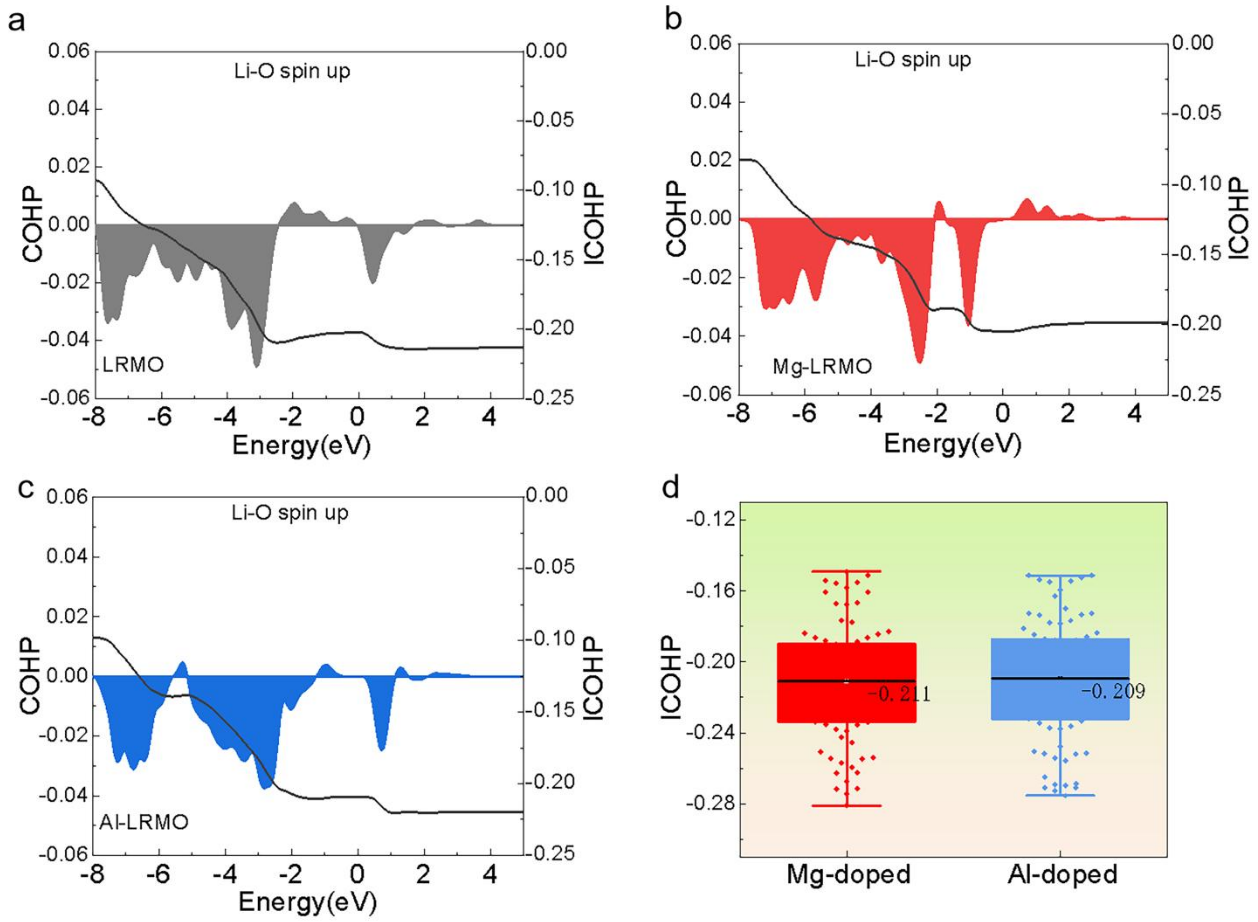

Figure S22. The COHP and ICOHP analysis of Li-O bonds spin-up for (a) LRMO, (b) Mg-LRMO, (c) Al-LRMO. (d) The ICOHP analysis of Li-O for the Mg-LRMO and Al-LRMO.

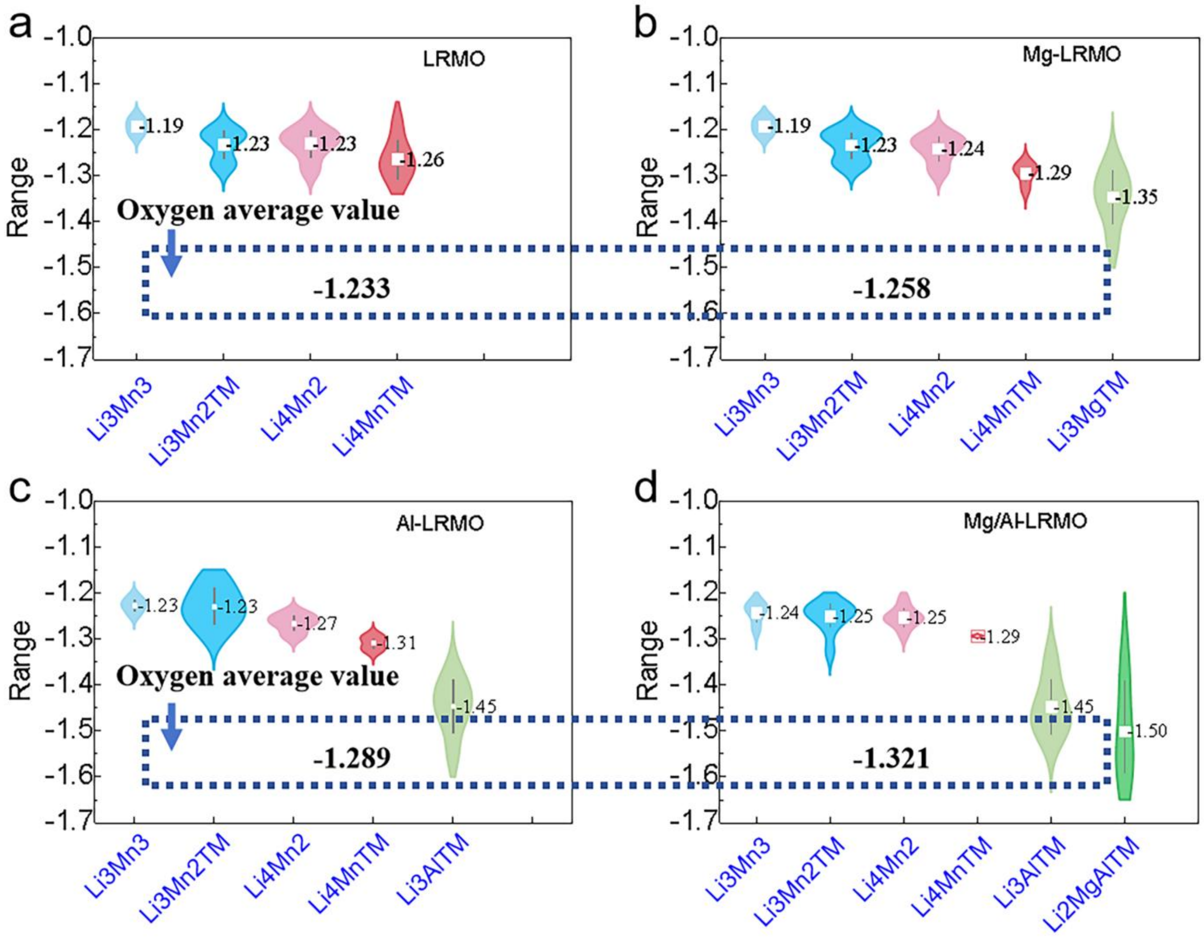

Figure S23. The average atomic charge of O anions with different cation coordination of (a) LRMO, (b) Mg-LRMO, (c) Al-LRMO and (d) Mg/Al-LRMO.

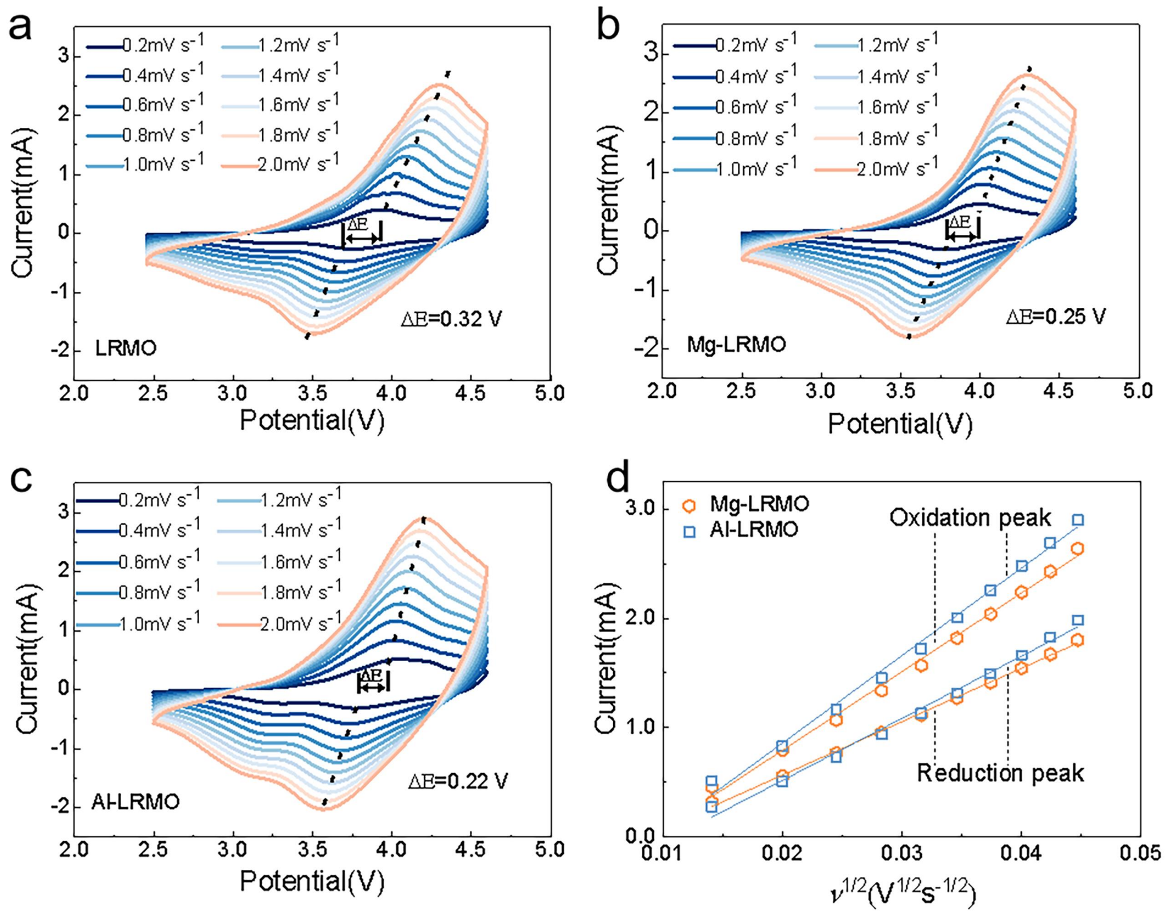

Figure S24. CV profiles of (a) LRMO, (b) Mg-LRMO, and (c) Al-LRMO at different scan rates between 2.5 and 4.6 V. (d) oxidation peak and reduction peak of Mg-LRMO, and Al-LRMO.

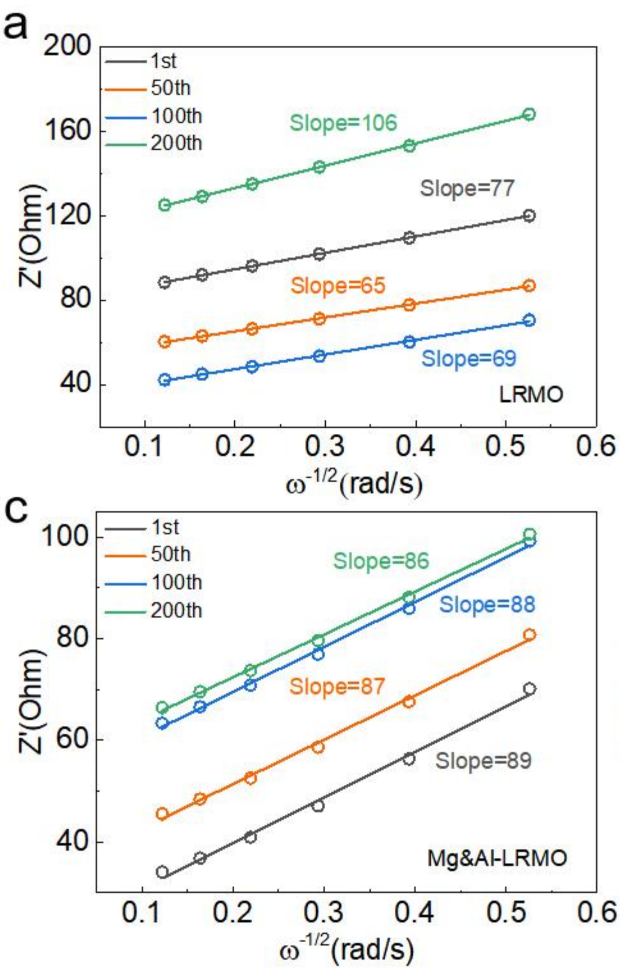

Figure S25. $\omega^{-1/2}$ relationship diagram and fitting result of the corresponding impedance from fitting result of EIS (a,b) LRMO and (c,d) Mg/Al-LRMO electrode.

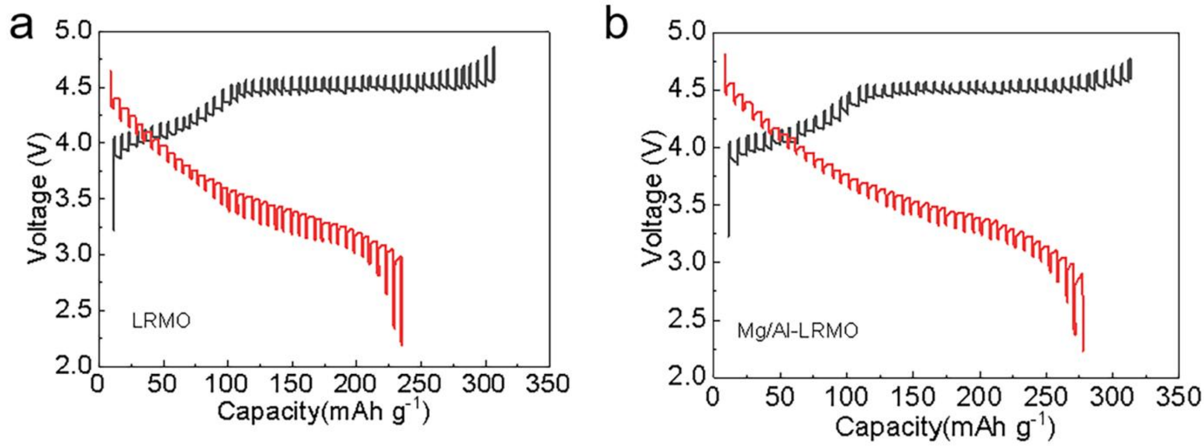

Figure S26. GITT curves of (a) LRMO and (b) Mg/Al-LRMO between 2.5 and 4.6 V.

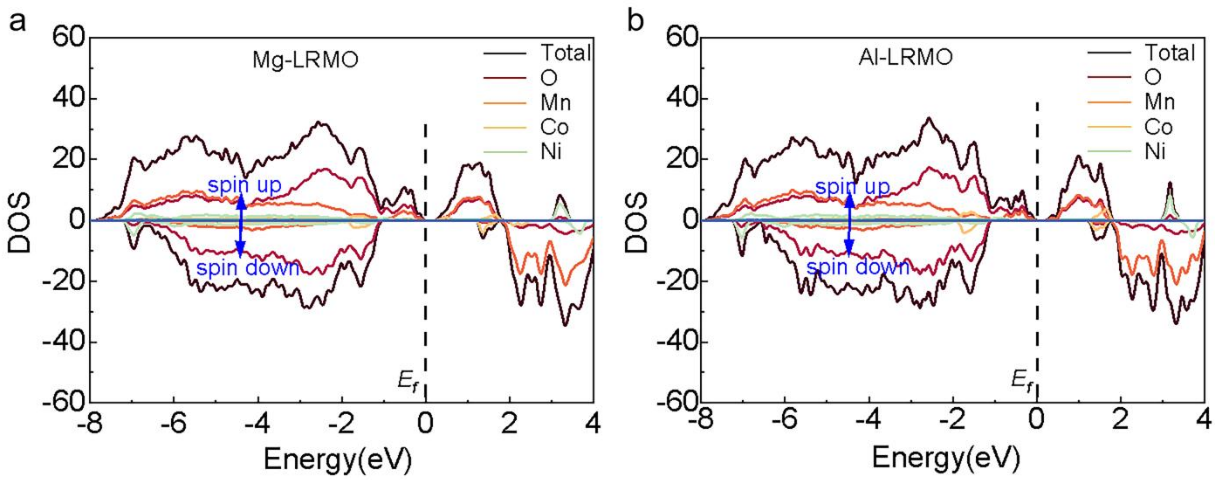

Figure S27. Total and Projected density of states (DOS) of (a) Mg-LRMO and (b) Al-LRMO (with the Fermi level set to 0 eV).

Supplementary Tables

Table S1. Chemical composition results of ICP-OES analysis of LRMO, Mg-LRMO, Al-LRMO and Mg/Al-LRMO.

Sample	Chemical composition (at%)					
	Li	Mn	Ni	Co	Mg	Al
LRMO	1.417	0.687	0.201	0.112	0	0
Mg-LRMO	1.408	0.686	0.200	0.111	0.012	0
Al-LRMO	1.414	0.683	0.197	0.108	0	0.012
Mg/Al-LRMO	1.403	0.681	0.198	0.109	0.013	0.011

Table S2. Refined lattice parameters of LRMO, Mg-LRMO, Al-LRMO and Mg/Al-LRMO.

Sample		LRMO	Mg-LRMO	Al-LRMO	Mg/Al-LRMO
R-3m phase	a (Å)	2.83625(3)	2.83731(3)	2.83754(3)	2.83976(3)
	c (Å)	14.11689(5)	14.16609(9)	14.17785(3)	14.29659(9)
	c/a	4.97731	4.99279	4.99653	5.03443
C2/m phase	a (Å)	4.94746(1)	4.94841(6)	4.94855(4)	4.94973(1)
	b (Å)	8.46204(2)	8.46382(1)	8.46534(0)	8.46625(7)
	c (Å)	5.18463(3)	5.22736(1)	5.22749(0)	5.22867(3)
$I_{(003)}/I_{(104)}$		1.34	1.56	1.63	1.88
Amount of Ni at Li Sites(%)		9.38	7.47	7.58	6.12
Percentage composition	*R-3m* phase	55.34%	48.59%	47.75%	44.38%
	C2/m phase	44.66%	51.41%	52.25%	55.62%
Li-O(Å)	*R-3m* phase	2.1027	2.1031	2.1033	2.1036
	C2/m phase	2.0736	2.0786	2.0793	2.0857
TM-O(Å)	*R-3m* phase	2.0137	2.0135	2.0134	2.0131
	C2/m phase	1.9571	1.9568	1.9561	1.9756
Z_{ox}		0.2558(5)	0.2586(2)	0.2588(7)	0.2594(3)
S_{MO2} (Å)		2.1876	2.1368	2.1215	2.1131
I_{LiO2} (Å)		2.5180	2.6052	2.6145	2.6524
Rp (%)		5.02	3.56	3.57	3.48
R_{wp}(%)		6.69	4.88	5.06	4.61

Table S3. Initial charge-discharge data of the LRMO, Mg-LRMO, Al-LRMO and Mg/Al-LRMO.

Sample	Charge capacity (mAh g^{-1})	Discharge capacity (mAh g^{-1})	Coulombic efficiency (%)	Sloping region (mAh g^{-1})	Plateau region (mAh g^{-1})
LRMO	329.2	237.7	72.2	110.2	219
Mg-LRMO	322.4	245.1	76.0	113.3	209.1
Al-LRMO	317.5	263.1	82.9	117.0	200.5
Mg/Al-LRMO	315.0	269.9	85.7	120.1	194.9

Table S4. The Mn, Ni, Co and O related area data of the LRMO, Mg-LRMO, Al-LRMO and Mg/Al-LRMO from XPS profiles.

Samples	Mn (%)		Ni (%)		Co (%)		O (%)		
	Mn^{3+}	Mn^{4+}	Ni^{2+}	Ni^{3+}	Co^{2+}	Co^{3+}	O^{2n-}	Oxygen Vacancies	Oxygen Lattice
LRMO	37.99	62.01	59.06	40.94	48.84	51.16	21.08	22.87	56.05
Mg-LRMO	27.45	72.55	44.88	55.12	42.65	57.35	14.94	26.24	58.82
Al-LRMO	22.24	77.76	38.12	61.82	39.19	60.81	12.38	27.15	60.47
Mg/Al-LRMO	20.28	79.72	27.77	72.23	36.25	63.75	6.11	32.69	61.20

Table S5. The comparisons of electrochemical properties between this work and reported representive modification studies on LLOs.

Modification	Voltage (V)	Current density (mA g^{-1})	Discharge capacity (mAh g^{-1})	Rate performance (mAh g^{-1}/C)	Cycle numbers	Capacity retention (%)	Voltage fading (mV/cycle)	Ref
Al doping	2.0-4.8	250	195/1C	120/5C	200(1C)	92.9	2.0mV/cycle	[9]
Al doping	2.0-4.7	250	125/1C	77/3C	250(0.2C)	67.0	2.3mV/cycle	[10]
Al doping	2.0-4.8	250	127/1C	78/5C	100(0.2C)	84.5	2.1mV/cycle	[11]
Al doping	2.0-4.8	250	173/1C	142/5C	150(0.1C)	92.5	2.4mV/cycle	[12]
Al doping	2.0-4.6	300	225/1C	130/4C	35(0.1C)	85.2	2.4mV/cycle	[13]
Al doping	2.0-4.8	250	166/1C	110/5C	100(0.5C)	96.4	5.0mV/cycle	[14]
Al doping	2.5-4.6	250	203/1C	142/5C	200 (1C)	80.6	1.0mV/cycle	**This work**
Mg doping	2.7-4.6	180	142/1C	125/3C	100(0.5C)	73.5	---	[15]
Mg doping	2.0-4.8	300	182/1C	119/4C	50(0.5C)	94.0	---	[16]
Mg doping	2.0-4.8	250	221/1C	147/3C	100(0.2C)	76.1	----	[4]
Mg doping	2.0-4.6	200	220/1C	172/5C	100(5.0C)	92.6	0.9mV/cycle	[17]
Mg doping	2.0-4.6	200	225/1C	175/5C	100(0.5C)	96.4	2.5mV/cycle	[18]
Mg doping	2.5-4.6	250	192/0.5C	170/1C	100(1C)	82.6	----	[19]

Mg doping	2.5-4.6	250	195/1C	116/5C	200 (1C)	76.3	0.8mV/cycle	**This work**
Al/F doping	2.0-4.8	250	225/1C	157/10C	150(0.5C)	88.2	4.0mV/cycle	[20]
Al/B doping	2.0-4.8	250	216/1C	200/2C	80(0.1C)	82.5	3.5mV/cycle	[21]
Al/Nb doping	2.0-4.7	250	205/1C	168/5C	200(1C)	82	3.8mV/cycle	[22]
Al/Cr doping	2.0-4.8	240	205(1C)	123/10C	100(0.2C)	97.3	3.8mV/cycle	[23]
Mg/La doping	2.0-4.8	200	204(1C)	121/5C	150(0.5C)	86.2	4.0mV/cycle	[24]
Mg/F doping	2.0-4.8	230	191(1C)	151/3C	100(1C)	88.6	1.2mV/cycle	[25]
Mg/Al doping	2.0-4.8	250	182(1C)	120/5C	100(0.1C)	81.6	4.9mV/cycle	[26]
Mg/Al doping	2.5-4.6	250	210 (1C)	160/5C	200 (1C)	90.0	0.7mV/cycle	**This work**

Table S6. Li^+ diffusion coefficient (D_{Li^+}) of the LRMO, Mg-LRMO, Al-LRMO and Mg/Al-LRMO from CV.

Samples	D_{Li^+} ($cm^2\ s^{-1}$) (Deintercalation)	D_{Li^+} ($cm^2\ s^{-1}$) (Intercalation)
LRMO	1.06×10^{-14}	1.56×10^{-14}
Mg-LRMO	1.03×10^{-14}	1.51×10^{-14}
Al-LRMO	9.23×10^{-13}	1.30×10^{-14}
Mg/Al-LRMO	8.38×10^{-13}	1.29×10^{-13}

Table S7. The impedance parameters and the diffusion coefficient (D_{Li^+}) of the LRMO and the Mg/Al-LRMO electrodes calculated from the equivalent circuit.

Samples	States	Rs(Ω)	R_{ct}(Ω)	D_{Li^+} ($cm^2\ s^{-1}$)
LRMO	before	3.26	31.09	6.59×10^{-14}
	20th	3.78	42.25	1.55×10^{-13}
	100th	4.07	63.16	1.75×10^{-13}
	200th	5.36	106.51	3.82×10^{-13}
Mg/Al-LRMO	before	2.28	24.1	9.34×10^{-14}
	20th	2.59	31.07	9.78×10^{-14}
	100th	3.76	40.45	9.56×10^{-14}
	200th	4.12	43.14	1.05×10^{-13}

References

[1] a) G. Kresse, J. Furthmüller, *Physical review. B, Condensed matter* **1996**, 54, 11169; b) K. Shimoda, T. Minato, K. Nakanishi, H. Komatsu, T. Matsunaga, H. Tanida, H. Arai, Y. Ukyo, Y. Uchimoto, Z. Ogumi, *Journal of Materials Chemistry A* **2016**, 4, 5909.

[2] P. E. Blöchl, *Physical Review B* **1994**, 50, 17953.

[3] a) C. Wang, S. Wang, Y.-B. He, L. Tang, C. Han, C. Yang, M. Wagemaker, B. Li, Q.-H. Yang, J.-K. Kim, F. Kang, *Chemistry of Materials* **2015**, 27, 5647; b) L. M. Milne-Thomson, *Nature* **1945**, 156, 190.

[4] C. Huang, Z. Wang, H. Wang, D. Huang, Y.-b. He, S.-X. Zhao, *Materials Today Energy* **2022**, 29, 101116.

[5] W. Choi, H.-C. Shin, J. M. Kim, J. Y. Choi, W. S. Yoon, *Journal of Electrochemical Science and Technology* **2020**, 11, 1.

[6] R. Yu, X. Wang, D. Wang, L. Ge, H. Shu, X. Yang, *Journal of Materials Chemistry A* **2015**, 3, 3120.

[7] M. Dong, Z. Wang, H. Li, H. Guo, X. Li, K. Shih, J. Wang, *ACS Sustainable Chemistry &Engineering* **2017**, 5, 10199.

[8] a) P. Zhang, X. Zhai, H. Huang, J. Zhou, X. Li, Y. He, Z. Guo, *Ceramics International* **2020**, 46, 24723; b) Y. Liu, X. Fan, X. Huang, D. Liu, A. Dou, M. Su, D. Chu, *Journal of Power Sources* **2018**, 403, 27.

[9] J. Yin, N. Wang, C. Lin, W. Liu, S.-T. Myung, Y. Jin, *Chemical Engineering Journal* **2023**, 474, 145552.

[10] D. De Sloovere, S. K. Mylavarapu, J. D'Haen, T. Thersleff, A. Jaworski, J. Grins, G. Svensson, R. K. Stoyanova, L. O. Jøsang, K. R. Prakasha, M. Merlo, E. Martínez, M. Nel-Lo Pascual, J. Jacas Biendicho, M. K. van Bael, A. Hardy, *Small* **2024**, e2400876.

[11] J.-Y. Wang, Y.-L. Xie, S.-M. Yang, *Journal of Electroanalytical Chemistry* **2024,** 118453.

[12] F. Yang, Y. Li, Z. Wang, C. Chen, Z. Guo, *Journal of Applied Electrochemistry* **2023**, 53, 1.

[13] J. Ma, B. Li, L. An, H. Wei, X. Wang, P. Yu, D. Xia, *Journal of Power Sources* **2015**, 277.

[14] Z. Shi, X. Wang, W. Feng, C. Song, L. Chen, M. Li, *Ionics* **2020**, 26, 5951.

[15] Z. Xiao, C. Zhou, L. Song, Z. Cao, P. Jiang, *Ionics* **2021**, 27, 1909.

[16] Y. X. Wang, K. H. Shang, W. He, X. P. Ai, Y. L. Cao, H. X. Yang, *ACS applied materials & interfaces* **2015**, 7, 13014.

[17] R. Yu, X. Wang, Y. Fu, L. Wang, S. Cai, M. Liu, B. Lu, G. Wang, D. Wang, Q. Ren, X. Yang, *Journal of Materials Chemistry A* **2016**, 4, 4941.

[18] R. Yu, G. Wang, X.-y. Wang, D. Wang, W. Wen, H. Shu, X. Yang, *ChemElectroChem* **2015**, 2, 1346.

[19] H. Ouyang, X. Li, Z. Wang, H. Guo, W. Peng, *Ionics* **2018**, 24, 3347.

[20] B. Guo, J. Zhao, X. Fan, W. Zhang, S. Li, Z. Yang, Z. Chen, W. Zhang, *Electrochimica Acta* **2017**, 236, 171.

[21] J. Huang, F. Zhu, Y. Shu, G. Hu, Z. Peng, Y. Cao, J. Jiang, S. Zhang, X. Wang, K. Du, *Ceramics International* **2021**, 47, 34611.

[22] W. Jiang, C. Zhang, Y. Feng, B. Wei, L. Chen, R. Zhang, D. G. Ivey, P. Wang, W. Wei, *Energy Storage Materials* **2020**, 32, 37.

[23] Y. Liu, X. Fan, Z. Zhang, H.-H. Wu, D. Liu, A. Dou, M. Su, Q. Zhang, D. Chu, *ACS Sustainable Chemistry & Engineering* **2019**, 7, 2225.

[24] M. Li, H. Wang, L. Zhao, Y. Zhou, F. Zhang, D. He, *Journal of Alloys and Compounds* **2019**, 782, 451.
[25] Z. Guo, L. Li, Z. Su, G. Peng, M. Qu, Y. Fu, H. Wang, W. Ge, *Electrochimica Acta* **2023**, 437, 141525.
[26] Y. Liang, S. Li, J. Xie, L. Yang, W. Li, C. Li, L. Ai, X. Fu, X. Cui, X. Shangguan, *New Journal of Chemistry* **2019**, 43, 12004.